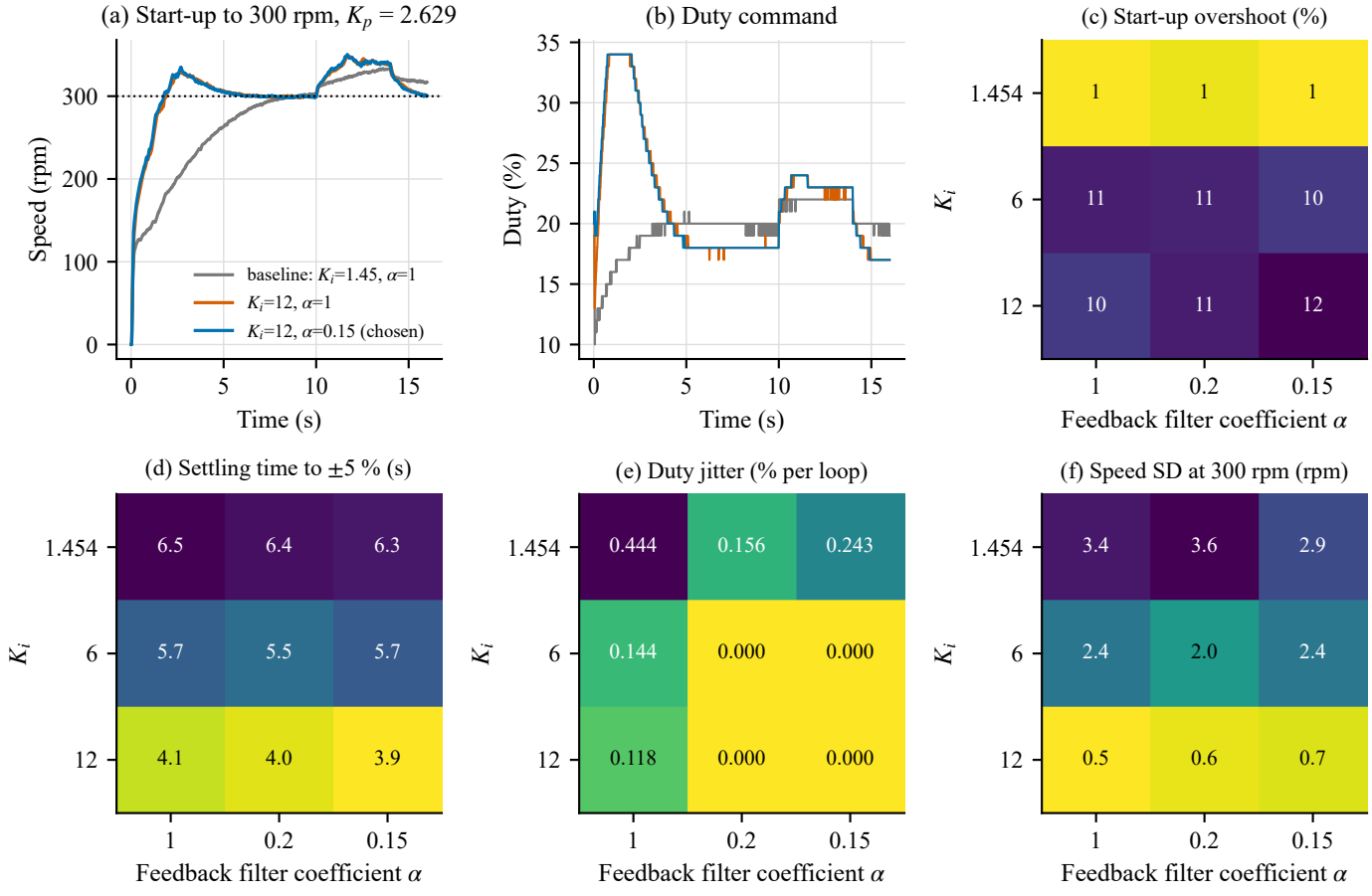

(a) Start-up to 300 rpm, K_p = 2.629
Speed (rpm)
Time (s)
baseline: K_i=1.45, α=1
K_i=12, α=1
K_i=12, α=0.15 (chosen)
(b) Duty command
Duty (%)
Time (s)
(c) Start-up overshoot (%)
1.454
6
12
1 1 1
11 11 10
10 11 12
K_i
1
0.2
0.15
Feedback filter coefficient α
(d) Settling time to ±5 % (s)
6.5 6.4 6.3
5.7 5.5 5.7
4.1 4.0 3.9
(e) Duty jitter (% per loop)
0.444 0.156 0.243
0.144 0.000 0.000
0.118 0.000 0.000
(f) Speed SD at 300 rpm (rpm)
3.4 3.6 2.9
2.4 2.0 2.4
0.5 0.6 0.7